Embodied Multimedia: When Multimedia Meets Embodied Intelligence

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Multimedia technology has historically focused on optimizing sensory experiences for human observers. However, with the paradigm shift from conventional Artificial Intelligence to Embodied Intelligence, intelligent systems are evolving into agents that actively perceive and interact with the physical world. The transition renders traditional multimedia frameworks that are designed for passive consumption and unidirectional transmission insufficient, as they lack the semantic understanding and real-time feedback required for embodied tasks. To bridge this gap, we formally propose **Embodied Multimedia**, providing a comprehensive review of this emerging paradigm and a unified four-layer architecture (Data, Communication, Cognitive, and Evaluation layers). Within the proposed framework, we systematically review the key enabling technologies that underpin embodied agents. Furthermore, we establish a multi-dimensional evaluation system covering perception, cognition, and action. Finally, we discuss the open challenges and future research directions to facilitate the deep integration of multimedia technologies with embodied intelligence.

CCS Concepts: • **Information systems** → **Multimedia information systems**; • **Computing methodologies** → **Artificial intelligence**; • **Computer systems organization** → *Robotics*.

Additional Key Words and Phrases: Embodied Multimedia, Embodied Intelligence, Multimodal Computing, Agent Learning

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1 INTRODUCTION

Multimedia technology serves as the central driver of the information age, seamlessly integrating diverse modalities such as text, images, audio, video, and 3D data [224]. Through the integration of varied modalities, multimedia computing has fundamentally reshaped how humans access information and perceive the world. From the early combination of graphics and text to modern immersive virtual and extended reality technologies, traditional multimedia systems have consistently adhered to a human-centric philosophy, aiming to optimize the sensory experience for human users. Consequently, decades of research have established a comprehensive framework for data acquisition, processing, transmission, and evaluation. However, as Artificial Intelligence (AI) transitions from the digital realm into complex physical environments, the rapid rise of Embodied Intelligence is initiating a profound technological disruption [113].

Unlike conventional AI systems that passively process pre-collected datasets in cyberspace, embodied systems are autonomous agents equipped with physical bodies that must continuously perceive, reason, and act in real-time within dynamic environments. Such active engagement demands a profound synergy of morphology, action, perception, and learning [105], aiming to effectively align cyberspace with the physical world [113]. Under the embodied framework, multimedia data is no longer merely content for human consumption; rather, it serves as the essential sensory foundation that bridges an agent's cognition and physical execution. Consequently, the paradigm shift from human-centric consumption to agent-centric active interaction exposes critical vulnerabilities in the traditional multimedia pipeline, creating a critical bottleneck for the real-world deployment of embodied agents.

Specifically, traditional multimedia technologies are severely misaligned with the requirements of embodied tasks, presenting formidable challenges across the entire processing chain. First, regarding data acquisition and processing, traditional systems rely on passive capture from fixed viewpoints and utilize compression standards mathematically formulated to discard high-frequency visual information imperceptible to the human eye [164]. However, for machine vision, the inability to actively resolve visual occlusions, coupled with the irreversible loss of fine-grained edges and textures, directly degrades the accuracy of downstream reasoning, potentially leading to severe physical interaction errors [88]. Second, traditional communication and retrieval methods prioritize reliable, unidirectional bit-stream transmission and static retrieval matching. Furthermore, traditional multimedia generative models are primarily designed to produce content that caters to human interests; they often lack an emphasis on generating physics-grounded outputs, making it difficult to support physical interactions in embodied multimedia [14]. Finally, traditional evaluation metrics merely quantify pixel-level signal fidelity [188]. They fail to effectively measure the decision-making value of data or reflect embodied task performance metrics like spatial reasoning accuracy, physical interaction safety, and the synergy of action and perception [119].

To address the aforementioned challenges, it has become essential to establish a new multimedia paradigm designed for embodied agents and tasks, known as Embodied Multimedia. In recent years, several reviews [37, 105, 113, 130, 144, 165, 168, 181, 207, 224] have explored related technologies across the Embodied Intelligence and Multimedia domains, generating various analytical frameworks. We analyze the core focal points covered in these recent reviews across dimensions such as multimodal perception, cognition, physical action, and system architecture, as shown in Table 1. Although existing studies have separately focused on specific modalities or partial control loops, there remains a critical lack of systematic review and architectural definition for this emerging cross-disciplinary field from a holistic multimedia perspective. To fill the gap, we introduce the concept of Embodied Multimedia for the first time, providing a comprehensive tutorial on its basic concepts and fundamental principles alongside a systematic review of the relevant literature. Unlike previous studies that focus on a single modality or specific algorithms, this survey proposes a unified

Table 1. Comparison of our survey against related Embodied AI and Multimedia and literature.

Reference	Domain	Core Scope & Focus	Multimodal Perception	Communication	Cognition & Learning	Physical Action	Sim-to-Real	Unified Architecture
[165]	Embodied AI	Perception-Cognition-Behavior (PCB) LLMs & generalist agents	●	–	●	●	●	–
[105]	Embodied AI	Synergy of agent morphology, action, perception and learning	●	–	●	●	◐	–
[144]	Embodied AI	Multi-sensor fusion perception & multi-agent in embodied AI	●	–	◐	–	◐	–
[113]	Embodied AI	Multimodal LLMs, world models & cyber-physical world alignment	●	–	●	●	●	–
[181]	Embodied AI	RL, motion planning & control for mobile robots	◐	–	●	●	●	–
[37]	Multimedia	Deep learning-based content-based image retrieval (CBIR)	●	–	◐	–	–	–
[130]	Multimedia	Survey of generative models for cross-modal visual content	●	–	●	◐	–	–
[224]	Multimedia	Methods, modality combinations & applications of multimodal ML	●	–	●	–	–	–
[168]	Multimedia	Video-LLMs for video understanding with multi-granularity reasoning	●	–	●	–	–	◐
[207]	Multimedia	Multimodal recommendation & implicit user preference mining	●	–	●	–	–	–
Ours	**Cross-domain**	**Unified Embodied Multimedia architecture (Data-Comm.-Cognition-Eval.)**	●	●	●	●	●	●

Symbol legend: ● Full coverage; ◐ Partial coverage; – Not covered

architecture for Embodied Multimedia. The architecture consists of the Data Layer, the Communication Layer, the Cognitive Layer, and the Evaluation Layer, covering the entire information processing workflow. Specifically, the primary contributions of this survey are summarized as follows:

- This survey systematically traces the evolution from traditional multimedia to Embodied Multimedia. We provide a detailed analysis of the limitations of traditional multimedia computing in data processing, system architecture, and evaluation systems within the context of the embodied intelligence era. Furthermore, we establish a core definition of Embodied Multimedia as the central link between perception and action and propose a unified four-layer Embodied Multimedia architecture.
- This survey provides a comprehensive and structured review of the critical technologies within the proposed four-layer architecture. Specifically, we discuss data acquisition, generation, and semantic compression in the Data Layer; task-oriented semantic communication and cloud-edge collaborative computing in the Communication Layer; core decision-making technologies (including agent-based retrieval, world models, Vision-Language-Navigation (VLN), and Vision-Language-Action (VLA)) in the Cognitive Layer; and establish a multi-dimensional assessment system covering perception, cognition, and action in the Evaluation Layer.
- This survey rigorously analyzes the technical challenges inherent in current architectures and identifies promising future research directions, aiming to provide a theoretical foundation and technical guidance for the deep integration of multimedia technology and Embodied Intelligence.

The overall structure of this survey is illustrated in Figure 1, and the remainder of this article is as follows: Section 2 details the developmental history of multimedia technology and the rise of Embodied Intelligence, providing an in-depth analysis of the limitations of traditional multimedia in embodied scenarios. Section 3 formally introduces the definition of Embodied Multimedia and its four-layer unified architecture. Sections 4 through 7 systematically discuss the key technologies and recent advancements within each specific layer: the Data Layer (Section 4), the Communication Layer (Section 5), the Cognitive Layer (Section 6), and the Evaluation Layer (Section 7). Section 8 explores the open challenges and future trends in Embodied Multimedia technology. Finally, Section 9 provides the conclusion.

2 BACKGROUND AND MOTIVATION

Multimedia computing is a core task of information technology. It integrates, processes, stores, transmits, and presents various forms of information, such as text, images, audio, video, animation, and 3D data. Consequently, it has profoundly transformed how humans access information and perceive the world.

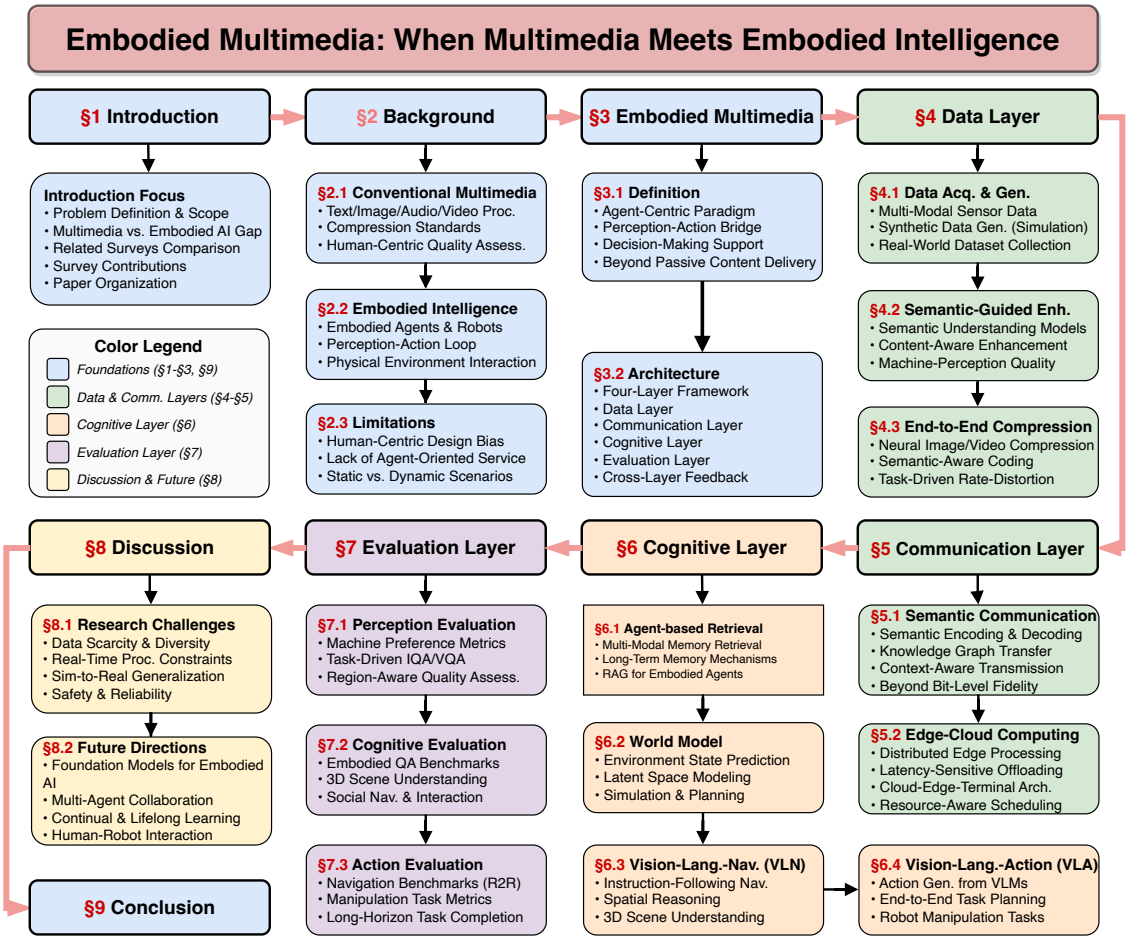

Fig. 1. Organization of this survey on Embodied Multimedia, where we organize into five dimensions: (1) Foundations (§1–§3, §9), (2) Data & Comunication Layers (§4–§5), (3) Cognitive Layer (§6), (4) Evalution Layer (§7), and (5) Discussion & Future (§8).

However, as AI moves from connectionism model design toward physical reality, the paradigm of Embodied Intelligence is rapidly developing. Under the new paradigm, multimedia data is no longer merely content for human consumption and appreciation; instead, it serves as the fundamental basis for agents to perceive the physical environment and guide their actions. The paradigm shift from human-centric to agent-centric services marks a new stage in the development of multimedia technology.

This section reviews the evolution of traditional multimedia and Embodied Intelligence, analyzing the inherent limitations of conventional systems in embodied scenarios. This analysis highlights the necessity of establishing a novel, agent-centric paradigm: Embodied Multimedia.

2.1 Conventional Multimedia

Traditional multimedia technology is driven by the goal of optimizing the Quality of Experience (QoE) for human users. By coordinating multi-modal information such as text, audio, images, and video, it has established a complete

technical chain ranging from data processing to immersive presentation. In terms of data compression, lossy compression techniques [164, 180, 196] achieve high-fidelity signal reconstruction within limited bandwidth by removing components that are imperceptible to human senses. For content retrieval, Content-Based Image Retrieval (CBIR) [159] utilizes algorithms like the Scale-Invariant Feature Transform (SIFT) [116] to enable efficient indexing of unstructured data. Regarding data transmission, Dynamic Adaptive Streaming over HTTP (DASH) [162] introduces bit-rate adaptation and buffer control mechanisms to ensure the continuity and low latency of audiovisual streams in heterogeneous network environments. In content generation, the introduction of Generative Adversarial Networks (GANs) [47] and Diffusion Models [160] has enabled the creation of diverse and rich images and videos. For display and interaction, the development of Ultra-High-Definition video, High Dynamic Range (HDR) [150], Virtual Reality (VR), and Augmented Reality (AR) [12] has provided users with more immersive sensory experiences. As display technology has advanced, evaluation systems have also expanded. Metrics have evolved from measuring 2D images and videos, such as PSNR, SSIM [188], FID [57], and FVD [138], to assessing omnidirectional image and video quality [167, 200, 209], and point cloud quality [23, 62, 110, 114, 169, 177, 237].

2.2 Embodied Intelligence

Embodied Intelligence is a paradigm that emphasizes how intelligent systems learn, adapt, and achieve intelligence through real-time and continuous interaction between their physical bodies and the environment. Unlike traditional AI systems, which focuses on symbolic processing, logical reasoning, or abstract data analysis, Embodied Intelligence posits that intelligence does not exist solely within a brain or a computational unit. Instead, it emerges from the dynamic interaction between a body and its environment. Existing research divides Embodied Intelligent systems into three parts: a perception module, a decision-making module, and an action module. The components together form a "perception-decision-action" closed-loop system [165], as illustrated in Fig 2. In the architecture, the perception component integrates multiple sensors—such as vision, touch, and inertial measurement—to achieve real-time sampling and feature extraction of multi-modal environmental states. The capability provides the system with dynamic environmental modeling. Based on the gathered environmental state, the decision-making module dynamically optimizes action strategies by considering the agent's internal state and goal constraints through a feedback regulation mechanism. Finally, the action component translates decision commands into physical action via actuators. The resulting impact on the environment is captured again by the perception component, forming a continuous, iterative loop. The feedback-based adaptive learning mechanism ensures that intelligence is not an isolated, static model but a comprehensive capability that emerges through the perception-decision-action cycle. The mechanism provides the theoretical support and implementation path for robust interaction in the complex physical world.

The development of Embodied Intelligence focuses on more efficient perception of the physical world, more accurate understanding of environmental semantics, and more robust execution of decision-making actions. In recent years, the interactive capabilities have been continuously enhanced and expanded due to the iterative progress of AI. During the initial stages of development, limited by computing power and algorithmic models, embodied agents were mostly restricted to specific tasks in closed environments. Their interaction patterns relied on preset rules or limited sensor inputs, making it difficult to handle the complex variables of the open world. However, the introduction of Large Multimodal Models (LMMs) has brought Embodied Intelligence into a new phase. Represented by Google's PaLM-E [35], embodied multimodal language models integrate visual encoders with Large Language Models (LLMs), enabling robots to understand complex physical common sense and perform multi-step reasoning tasks. At the same time, the scale and form of data for Embodied Intelligence have experienced explosive growth. To address the high cost of data

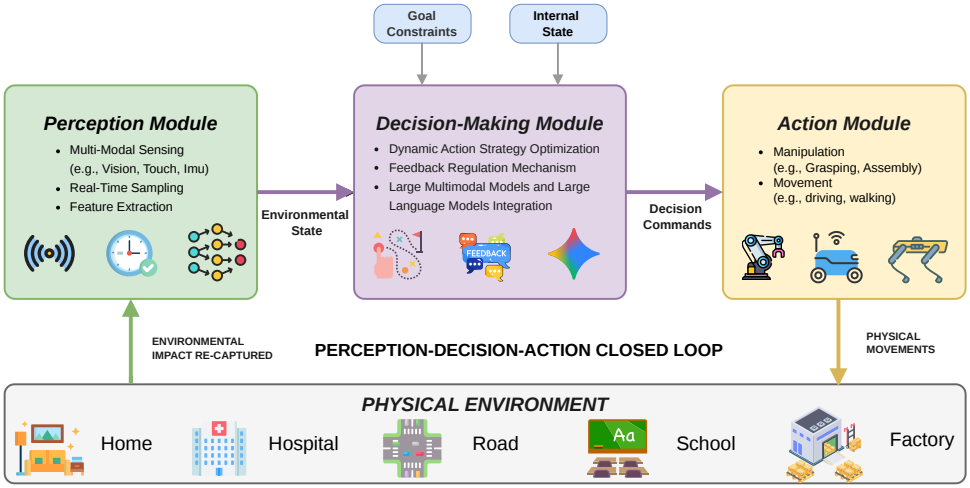

Fig. 2. Architecture of the perception-decision-action closed-loop system in Embodied Intelligence.

collection in the real world, a mainstream paradigm has emerged: large-scale synthetic data training based on physical simulation engines, such as Habitat [146], or Isaac Gym [124], combined with Sim-to-Real transfer technology. At the hardware level, bimanual robots [44, 239] and humanoid robots [28, 245] are undergoing rapid iteration. While updates in algorithms and hardware have pushed Embodied Intelligence toward general-purpose applications, existing multimedia information processing architectures show limitations in supporting their efficient operation. Therefore, there is an urgent need to evaluate and update the current technical framework. Figure 3 provides a comprehensive overview of the research landscape in Embodied Intelligence.

2.3 Limitations of Traditional Multimedia in the Embodied Intelligence Era

Although multimedia technology has established mature systems over the past decades, its primary objective remains the optimization of the sensory experience for human observers. However, as the service target shifts from human users to embodied agents that must actively perceive, reason, and interact within the physical environment, the traditional multimedia framework reveals significant limitations across its entire pipeline. The limitations, spanning data acquisition, communication, retrieval, generative modeling, and evaluation, necessitate a paradigm shift.

- **Limitations in Data Acquisition and Processing:** Traditional multimedia systems typically rely on passive data collection from fixed or pre-determined viewpoints, focusing on global scene capture. The static paradigm restricts an agent's ability to dynamically alter its perspective to resolve occlusions or actively explore unknown semantic regions [9, 24]. Furthermore, conventional image and video compression standards, such as JPEG [180] and HEVC [164], are fundamentally built upon human visual system models. They achieve compression by discarding high-frequency components that are imperceptible to humans. However, the process irreparably damages the fine-grained edge and texture features that are critical for machine vision, thereby compromising the reasoning accuracy of downstream embodied tasks [88].
- **Limitations in Communication Architectures:** The traditional communication paradigm is heavily grounded in Shannon's information theory, striving for bit-level reliable transmission of independent multimedia streams.

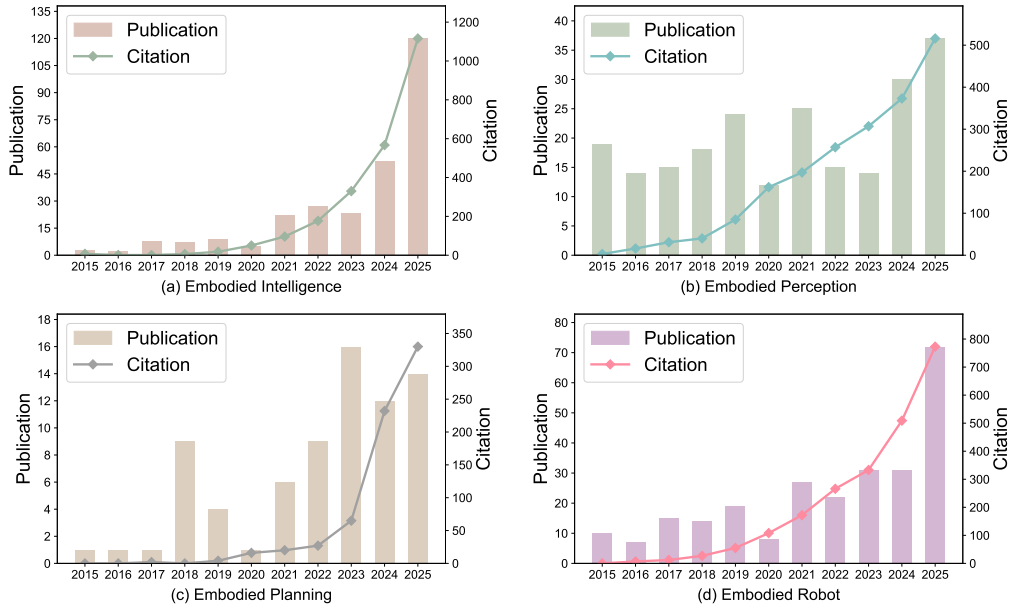

Fig. 3. Publication trends and citation analysis of research in Embodied Intelligence from 2015 to 2025.

In embodied scenarios, the unidirectional architecture becomes a bottleneck. It struggles to meet the rigorous demands for millisecond-level spatio-temporal alignment of heterogeneous multimodal data during real-time physical interactions. Moreover, relying solely on centralized cloud computing to process massive multi-sensory data streams incurs prohibitive latency. Existing communication frameworks and edge devices are inadequate for the massive computational demands and strict real-time control constraints required by large embodied foundation models [14, 90].

- **Limitations in Traditional Retrieval Technologies:** Traditional multimedia retrieval predominantly adheres to a static matching mechanism, constructing a unified feature representation space to calculate similarities between queries and heterogeneous data [38, 140]. While effective for explicit human queries, the static paradigms are inadequate for embodied agents facing long-sequence and knowledge-intensive tasks in open environments. Embodied tasks often involve ambiguous instructions that require dynamic context awareness and multi-step reasoning. Traditional systems lack a closed-loop mechanism for autonomous task analysis, sequential decision-making in retrieval, and feedback-driven optimization. [86, 159]. Consequently, they cannot proactively supply the adaptive, structured knowledge required for an agent's continuous cognitive processes.
- **Limitations in Traditional Generative Models:** Historically, multimedia generation has predominantly served human consumption, aiming to produce aesthetically pleasing text, images, and videos (e.g., traditional GANs [47] and latent diffusion models [143]). While the aforementioned models excel at creating digital content, they lack a deep understanding of physical laws, geometry, and affordances. Embodied agents, conversely, require generative models that can produce actionable outputs, such as predicting physically plausible future environmental dynamics [95] or directly outputting fundamental robotic control tokens [242]. Traditional generative models, confined to the digital realm, cannot effectively bridge the semantic gap to the physical action space.

Table 2. Comparison of conventional multimedia and Embodied Multimedia.

Dimension	Traditional Multimedia	Embodied Multimedia
Target User	Human observers	Embodied Intelligence
Core Goal	Optimises sensory experience and subjective perceptual fidelity	Optimises task execution efficiency and dynamic physical interaction
Data Acquisition	Passive data collection from fixed viewpoints, focusing on global scene capture	Active multimodal sensing with dynamic perspective adjustments and semantic exploration
Data Processing	Discards imperceptible high-frequency components to achieve visual compression	Retains critical semantic features via end-to-end compression and guided enhancement
Communication	Relies on bit-level reliable transmission and centralized cloud computing	Utilizes task-oriented semantic communication and edge-cloud collaborative computing
Retrieval Paradigm	Employs static matching in a unified feature space for explicit human queries	Leverages agent-based dynamic retrieval for autonomous task analysis and reasoning
Generative Modeling	Produces aesthetically pleasing digital content lacking deep physical understanding	Generates actionable outputs bridging semantics to physical manipulation and dynamics
Evaluation Metrics	Quantifies signal fidelity and human visual quality, such as PSNR and SSIM	Evaluates machine preference, task completion rates, and physical interaction safety

- **Limitations in Evaluation Metrics:** The evaluation systems entrenched in traditional multimedia are misaligned with the objectives of embodied tasks. Metrics such as PSNR and SSIM [188] primarily quantify signal fidelity to cater to QoE. However, these metrics fail to reflect the utility of the data for machine perception [88]. In embodied intelligence, visually distorted data might still perfectly support a successful robotic operation if critical semantic features are preserved. Crucially, traditional multimedia evaluation lacks embodied dimensions, such as long-horizon task completion rates, spatial reasoning accuracy, and physical interaction safety under hazardous instructions [57, 188].

Consequently, the profound mismatch across data, communication, cognition, and evaluation highlights the urgent need to establish a machine-and task-centered paradigm: Embodied Multimedia.

3 EMBODIED MULTIMEDIA

This section first defines Embodied Multimedia and then proposes a unified architecture for the field. The proposed architecture consists of four core layers: the Data Layer, the Communication Layer, the Cognitive Layer, and the Evaluation Layer. Through coordination and feedback between these levels, the architecture supports the continuous development and advancement of Embodied Multimedia.

3.1 Definition of Embodied Multimedia

Embodied Multimedia refers to a multimedia computing paradigm designed specifically for Embodied Intelligence. Unlike traditional human-centric multimedia, Embodied Multimedia treats multimodal data as a bridge connecting perception and action across the entire action loop. Furthermore, It also involves establishing evaluation metrics for perception, cognition, and action, enabling agents to achieve precise decision-making and continuous evolution within dynamic physical environments.

3.2 Architecture of Embodied Multimedia

To address the complex requirements of perception, decision-making, and action for Embodied Intelligence in dynamic physical environments, this survey proposes a four-layer Embodied Multimedia architecture. The architecture consists

of the Data Layer, the Communication Layer, the Cognitive Layer, and the Evaluation Layer. Together, the layers support the embodied agent in perceiving, deciding, and acting within the physical world.

Data Layer serves as the perceptual starting point of the Embodied Multimedia system, responsible for high-quality multimodal information acquisition and preprocessing. The Data Layer emphasizes adaptive strategies and collaborative multimodal sensing. Meanwhile, it introduces digital human motion generation technology to assist model training by synthesizing high-quality interaction data. Furthermore, the layer focuses on using semantic-guided enhancement to restore critical semantic regions. Finally, end-to-end compression technology is employed to alleviate the overhead of data transmission and storage, facilitating subsequent processing stages.

Communication Layer constitutes the neural pathway connecting the edge perception end to the cloud cognitive center. To meet the strict requirements of Embodied Intelligence for low latency and high reliability, the Communication Layer adopts the semantic communication paradigm. By extracting and encoding semantic features highly relevant to task intentions, the Communication Layer ensures the high-fidelity transmission of control commands and core environmental information, even in constrained or adverse network environments characterized by limited bandwidth or channel interference. Additionally, the Communication Layer introduces an edge-cloud collaborative computing mechanism. By pre-computing vision or motion prediction results at edge nodes, it reduces central latency and achieves real-time response.

Cognitive Layer is the decision-making core and information processing hub of the embodied agent, responsible for knowledge retrieval, environmental reasoning, spatial navigation, and motion generation. First, through an agent-based retrieval mechanism, the Cognitive Layer actively calls external multimedia knowledge bases, providing the agent with the knowledge transfer capability to handle long-term tasks and open-world scenarios. Regarding decision generation, the Cognitive Layer utilizes world models to construct dynamic representations of the environment, performing causal reasoning and "imagination" of the future states of the physical world to assist in complex planning. Meanwhile, to address the mobility requirements of embodied agents in unstructured environments, a VLN mechanism is introduced to achieve path planning and navigation decisions from natural language instructions to visual panoramic perception. Furthermore, by relying on VLA models, the layer achieves deep alignment between high-level semantic space and low-level physical action space, completing the direct mapping from multimodal input to physical control.

Evaluation Layer aims to establish a cross-layer global assessment model, encompassing perception evaluation, cognitive evaluation, and action evaluation. First, regarding perception evaluation, the layer quantifies the performance consistency of multimedia data in downstream tasks—such as object detection and semantic segmentation—by constructing machine preference databases and regional perceptual quality assessment models. The approach allows for the precise definition of the perceptual requirements for machine vision. Second, in terms of cognitive evaluation, the Evaluation Layer achieves multidimensional coverage ranging from spatial understanding and interaction planning to safety assessment. Finally, for action evaluation, the Evaluation Layer focuses on the mapping efficiency from high-level semantic instructions to low-level physical execution. Leveraging large-scale navigation and manipulation benchmarks, it rigorously evaluates the performance of agents in long-sequence tasks, open-vocabulary operations, and migration across different hardware platforms. By providing a comprehensive assessment of perception, cognition, and action, the layer offers a rigorous quantitative basis for model iteration and architectural optimization.

4 DATA LAYER

The Data Layer serves as the fundamental basis for embodied agents to perceive the physical environment. The section focuses on three key technologies: data acquisition and generation, semantic-guided enhancement, and end-to-end compression. It systematically explains how to optimize the agent's ability to understand complex dynamic environments starting from the information source.

4.1 Data Acquisition and Generation

To endow agents with efficient perception capabilities in dynamic environments, Embodied Multimedia introduces adaptive acquisition strategies and tactile sensing mechanisms at the data acquisition layer. Additionally, digital human motion generation technology is incorporated to mitigate the challenge of data scarcity.

Adaptive Acquisition: Unlike passive collection methods in traditional multimedia systems where sensor positions are relatively fixed, Embodied Multimedia enables systems to actively regulate collection and perception strategies based on task requirements—a paradigm known as adaptive acquisition. Historically, Bajcsy proposed the "Active Perception" theory in a foundational work [9], a perspective subsequently validated mathematically by Aloimonos et al. [2]. Specifically, Embodied Multimedia frameworks can dynamically calculate the next acquisition viewpoint that maximizes information gain, actively mitigating perceptual occlusions [24]. Furthermore, these systems can determine when to sample and which sampling strategies to employ. For instance, event cameras [45] can be utilized to collect data, significantly reducing redundancy in dynamic scenes. Additionally, an active exploration framework based on semantic curiosity [20] allows agents to avoid indiscriminate pixel collection when constructing environmental representations; instead, they selectively explore unknown semantic regions, substantially optimizing the task relevance of data acquisition.

Tactile Information Acquisition: To overcome single-modality limitations, the Data Layer integrates tactile sensing with vision and audition, emphasizing cross-modal complementarity and feature alignment. Tactile sensors are categorized into piezoresistive, capacitive, and vision-based types. In piezoresistive sensing, recent advances include using elastic nanofilms to resolve material-level mechanical mismatches [73], 3D micro-strain gauge arrays for hysteresis-free force decoupling [206], and localized gradient conductivity mechanisms to balance sensitivity with linear range [69]. Capacitive sensing research focuses on spatial and algorithmic enhancements, achieving non-contact and physical contact decoupling through topological optimization [102] and leveraging deep learning to resolve texture friction and normal pressure [112]. Vision-based sensors [34, 84, 192, 222] have gained prominence for their high resolution and low cost; frameworks like AnyTouch [39] further address sensor heterogeneity through unified representation learning. To facilitate task execution, multimodal systems such as Robot Synesthesia [223] and ViTaM [71] effectively fuse tactile data with 3D point clouds or visual information to reconstruct occluded deformations and enhance environmental comprehension.

Digital Human Motion Generation: Digital human motion generation aims to synthesize realistic, diverse, and semantically consistent human movements, serving as a pivotal technology for addressing the scarcity of training data for embodied agents. Figure 4 illustrates two representative motion generation paradigms. In the realm of diffusion models, MDM [170] pioneered the use of classifier-free diffusion models to predict motion samples combined with geometric losses, achieving high-quality text-driven generation. ReMoDiffuse [232] optimized the denoising process through a retrieval-augmented mechanism. For interaction scenarios, InterGen, proposed by Liang et al. [96], focuses on modeling multi-person interactions. OMOMO by Li et al. [94] and HOI-Diff by Peng et al. [133] utilize object motion guidance and the Affordance Prediction Diffusion Model (APDM), respectively. The methods not only achieve precise

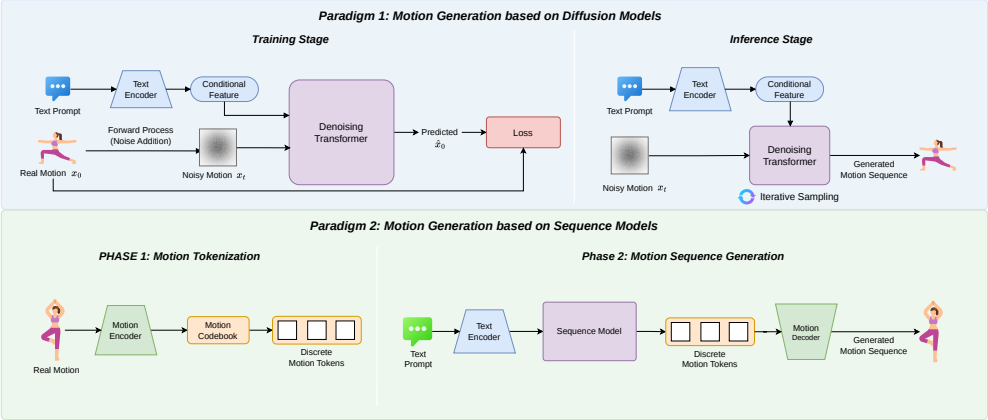

Fig. 4. Illustration of two representative digital human motion generation paradigms.

human-object interaction but also rectify physical artifacts through contact point constraints. Regarding sequence models, T2M-GPT [228] and MotionGPT [70] introduced Vector Quantized-Variational Autoencoders (VQ-VAE) and the GPT paradigm. The approaches discretize human actions into tokens for modeling, significantly improving the semantic consistency of generated motions. To further overcome bottlenecks in long-sequence generation and detail fidelity, MoMask [50] utilizes hierarchical masked modeling to achieve high-quality text-driven motion generation and high-fidelity motion completion. Meanwhile, Motion Mamba [236] designs a hierarchical temporal structure based on State Space Models (SSM), greatly enhancing the efficiency of long-sequence motion generation.

4.2 Semantic-Guided Enhancement

Multimedia enhancement primarily addresses degradation issues introduced during image and video acquisition and transmission, including Super-Resolution (SR), Denoising, and Deblurring. In the field of super-resolution, early classical methods based on prediction, edges, and sparse representation [189] have evolved into deep learning-based architectures. The approaches encompass CNN-based methods [33, 152], GAN-based methods [85, 183, 184], and Transformer-based methods [25, 30, 87, 97, 120]. In image denoising, the research focus has shifted from classical algorithms based on filtering, models, and partial differential equations [171] toward deep networks such as DnCNN [230], FFDNet [231], and RDUnet [51]. Similarly, deblurring technology has transitioned from blind deblurring using inverse filtering [148, 210] and regularization [8] to deep neural networks. Diverse approaches have emerged, including those based on Deep Auto-Encoders (DAE) [163], GANs [81, 82], multi-scale cascaded networks [129, 149], and re-blurring frameworks [229].

Traditional enhancement methods do not prioritize semantic features; instead, they perform uniform global enhancement on input images and videos, with optimization goals primarily centered on pixel-level fidelity. As the service target shifts to embodied agents, the objective of enhancement evolves to improving task efficiency and success rates. Consequently, enhancement in Embodied Multimedia has advanced into semantic-guided enhancement by integrating environmental semantic understanding and multimodal information fusion.

Semantic-Guided Enhancement: Recent studies have begun exploring the use of rich semantic priors embedded in SAM [234], CLIP [99, 121, 212], and generative diffusion models [166, 199] to provide semantic guidance for enhancement. For instance, to address the uneven distribution of computational resources in traditional methods, [234] achieves efficient

Table 3. Summary of representative semantic-guided multimedia enhancement methods and paradigms.

Year	Ref	Type	Prior Source	Contribution
2024	[234]	Restoration	SAM	Proposing a semantic distillation framework that transfers segmentation priors to lightweight networks for efficient restoration.
2023	[99]	Enhancement	CLIP	Utilizing the cross-modal capabilities of CLIP to address the challenge of non-uniform enhancement in backlit images.
2023	[121]	Restoration	CLIP	Utilizing vision-language alignment to train degradation-aware controllers for perceiving and repairing specific damage types.
2024	[212]	Deblurring	CLIP	Implementing unsupervised dual-pixel blur kernel estimation by exploiting the rich semantic space of CLIP.
2024	[166]	Super-Res	T2I Diffusion	Introducing cognitive embeddings as generative priors to ensure texture recovery follows the semantic logic of the real world.
2024	[199]	Super-Res	T2I Diffusion	Leveraging semantic prompts and diffusion priors to provide high-fidelity inputs for downstream recognition tasks.
2023	[202]	Enhancement	Semantic Knowledge Bank, SKB	Applying a semantic-aware knowledge-guided framework to ensure object-level adaptive brightness recovery via segmentation priors.

restoration of critical semantic regions by transferring segmentation priors from large models to lightweight networks. The DA-CLIP [121] framework leverages the vision-language alignment capability of CLIP to train a degradation-aware controller, empowering the model to discern the type of image damage and execute targeted restoration. CoSeR [166] and SeeSR [199] introduce Text-to-Image diffusion models as generative priors. By utilizing semantic prompts and cognitive embeddings, these models follow the semantic logic of the physical world when restoring textures, thereby providing high-confidence inputs for downstream recognition tasks. In addition to these methods, the Semantic-aware Knowledge-guided Framework (SKF) [202] ensures adaptive brightness restoration for different objects through semantic segmentation priors. The aforementioned methods no longer employ simple global enhancement for images or videos. Instead, they focus on restoring environmental geometric structures and object features that facilitate machine understanding, marking a paradigm shift for enhancement technology centered on semantic cognition. Table 3 summarizes the key semantic-guided enhancement methods discussed above.

4.3 End-to-End Compression

Data compression is a critical component in Embodied Multimedia systems, essential for reducing transmission bandwidth and improving storage efficiency. In recent years, end-to-end compression technologies based on deep learning have advanced rapidly, progressively outperforming traditional hybrid coding standards in terms of compression efficiency and reconstruction quality. This section provides an in-depth discussion covering two major domains: end-to-end image compression and end-to-end video compression.

End-to-End Image Compression: Image compression lays the theoretical groundwork for video compression. Early research explored variable-rate frameworks based on Recurrent Neural Network (RNN) [174]. Subsequently, Ballé et al. [10] established the mainstream architecture based on Variational Autoencoders (VAE), setting a cornerstone for the field through nonlinear transforms and uniform quantization. To address visual quality issues at extremely low bitrates, GANs were introduced into the compression pipeline, significantly enhancing the realism of reconstructed textures [1]. Regarding entropy coding optimization, the checkerboard context model [56] alleviated the serial bottleneck of autoregressive models. Furthermore, by integrating mixture attention blocks with the Swin Transformer [235], the capability for channel feature modeling was further strengthened.

End-to-End Video Compression: Early end-to-end video compression primarily imitated traditional hybrid coding frameworks. Rippel et al. [141] first explored learned video compression frameworks and low-latency coding schemes,

Table 4. Summary of representative end-to-end image and video compression methods.

Year	Ref	Type	Backbone	Contribution
2015	[174]	Image	RNN	Supporting variable-rate image compression via recurrent refinement and providing progressive reconstruction capabilities.
2016	[10]	Image	VAE	Establishing the VAE-based framework with non-linear transforms and uniform quantization to optimize rate-distortion performance.
2019	[1]	Image	GAN	Utilizing adversarial learning to synthesize textures at extremely low bitrates, achieving superior perceptual quality for environment perception.
2021	[56]	Image	CNN	Introducing a parallelizable checkerboard context model to eliminate the serial decoding bottleneck of autoregressive models.
2025	[235]	Image	Transformer	Enhancing entropy modeling through hybrid attention and Swin Transformer blocks to capture complex spatial dependencies.
2019	[141]	Video	CNN	Proposing a learned low-latency video compression pipeline that supports complex motion compensation and state maintenance.
2019	[117]	Video	CNN	First fully end-to-end video compression framework that jointly optimizes motion estimation, compensation, and residual coding.
2020	[106]	Video	CNN	Exploiting temporal correlation via first-order optical flow and second-order flow prediction within a ConvLSTM-based architecture.
2020	[100]	Video	CNN	Extending single-frame references to multi-frame prediction to enhance temporal correlation and reduce residual information.
2020	[83]	Video	CNN	Introducing pixel-level mode selection to adaptively choose between prediction and transmission for diverse motion scenarios.
2021	[61]	Video	CNN	Shifting motion compensation to the feature space using deformable convolutions to reduce pixel-domain artifacts and distortions.
2021	[91]	Video	CNN	Leading the paradigm shift from residual coding to conditional coding by leveraging high-dimensional feature-domain contexts.
2022	[151]	Video	CNN	Mining multi-scale temporal contexts from propagated features to improve compression efficiency for practical decoding times.
2022	[92]	Video	CNN	Combining spatial and temporal contexts with content-adaptive quantization to achieve significant bitrate savings over VTM/H.266.
2023	[93]	Video	CNN	Boosting performance by increasing context diversity in both temporal and spatial dimensions through hierarchical patterns and quadtree partitions.
2020	[213]	Video	RNN	Implementing a hierarchical quality layer structure and recurrent enhancement to improve reconstruction across video frames.
2020	[214]	Video	RNN	Utilizing recurrent auto-encoders and probability models to exploit long-range temporal dependencies and reduce bitrates.
2022	[126]	Video	Transformer	Directly modeling long-range dependencies with Transformers, effectively handling complex motion without explicit optical flow modules.
2022	[203]	Video	Transformer	Improving entropy coding through masked image modeling to capture multi-directional spatial-temporal dependencies.
2024	[178]	Video	CNN	Enabling the first real-time 1080p neural video decoding on mobile devices via integer quantization and hardware-accelerated warping.

while DVC [117] was the first to completely replace key modules in traditional video coding with neural networks. Building upon the foundation, to capture spatiotemporal correlations more accurately, optical flow prediction was used to mine temporal dependencies [106]. M-LVC [100] extended single-frame references to multi-frame references, and a mode selection mechanism [83] enhanced the model's adaptability to different motion scenarios. To overcome the bottlenecks of pixel-domain residual coding, research focus gradually shifted toward feature space and conditional coding. The FVC framework [61] shifted operations such as motion compensation into the feature space, reducing artifacts caused by pixel-level operations. DCVC [91] abandoned traditional residual coding in favor of conditional coding, utilizing feature-domain context to predict the distribution of the current frame. Subsequent studies further introduced temporal context mining [151], hybrid spatial-temporal entropy models [92], and diverse context mechanisms [93], significantly improving compression efficiency. Based on RNNs, HLVC [213] and RLVC [214] effectively utilized accumulated temporal information. The introduction of the Transformer architecture brought new breakthroughs to the field; the VCT model [126] utilized Transformers to handle long-sequence dependencies, completely removing explicit optical flow modules, while MIMT [203] further optimized entropy coding. Additionally, to meet the mobile deployment requirements of embodied agents, Rozendaal et al. [178] proposed MobileNVC, which achieved real-time decoding of 1080p video, demonstrating the engineering feasibility of end-to-end compression. Table 4 summarizes the key end-to-end image and video compression methods discussed above.

5 COMMUNICATION LAYER

The Communication Layer serves as the conduit connecting the edge perception end to the cloud cognitive center. To address the stringent requirements of Embodied Intelligence for low latency and high reliability, this section elaborates on a new task-oriented paradigm of semantic communication and explores the edge-cloud collaborative computing architecture. The discussion aims to elucidate how semantic transmission and collaborative computing can overcome

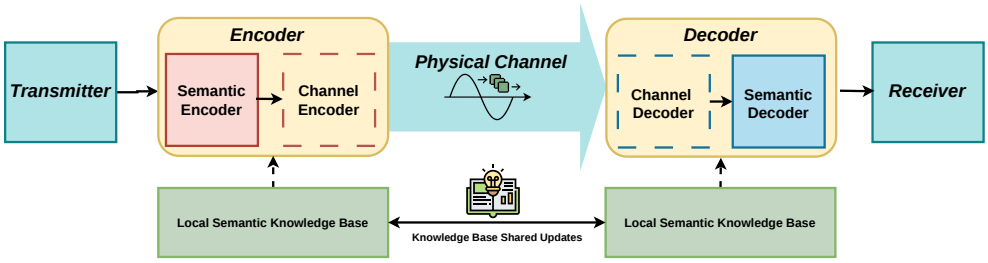

Fig. 5. Semantic communication architecture supported by a shared knowledge base.

bandwidth resource constraints and computational latency bottlenecks, ensuring the real-time response capabilities of agents in dynamic environments.

5.1 Semantic Communication

Semantic communication emerges as a transformative paradigm that shifts the fundamental objective of communication from bit-level reliable transmission to the preservation of underlying semantic meaning and communicative intent. From a structural perspective, semantic communication typically integrates semantic processing capabilities into conventional physical layer transmission mechanisms, though its implementation varies across different frameworks. In certain architectures, the semantic layer operates as a discrete module comprising a semantic encoder, a semantic decoder, and a shared knowledge base, where semantic features are extracted prior to conventional channel coding and reconstructed at the receiver through semantic inference guided by shared knowledge. Conversely, alternative architectures employ joint designs for semantic and channel coding. These frameworks typically utilize end-to-end deep learning models to directly map semantics to channel signals, thereby eliminating the explicit distinction between traditional coding and decoding modules [233]. Depending on the information modalities and application scenarios, research in the field primarily encompasses text-based semantic communication, speech or audio semantic communication, and image or multimodal semantic communication.

Textual Semantic Communication: In the field of textual semantic communication, DeepSC [205] employs attention mechanisms to extract deep semantic features from text, aiming to maximize system capacity and minimize semantic errors. To further optimize transmission quality, IS-JSCC [216] iteratively eliminates semantic errors caused by channel noise at the receiver. Addressing the issue of packet loss in semantic transmission, IK-HARQ [244] reuses the encoder during retransmission and utilizes incremental information for semantic error correction. Meanwhile, the SC-RS-HARQ architecture [72], which integrates traditional RS coding, introduces word error rate and sentence error rate optimization to further ensure semantic reliability.

Audio Semantic Communication: In the field of audio semantic communication, the DeepSC-S system [195] for speech transmission employs a CNN-based encoder-decoder architecture with attention mechanisms. By utilizing SE modules to extract key acoustic features, it can restore speech signals with high fidelity under dynamic channel conditions. Additionally, Tong et al. designed a CNN-based audio semantic autoencoder [175] and incorporated federated learning to train the model across distributed networks. The approach not only overcomes audio spectrum limitations but also effectively reduces transmission overhead.

Image or Multimodal Semantic Communication: For image data, an image semantic coding system built using GANs [63] employs a coarse-to-fine coding strategy. NTSCC [31] utilizes nonlinear transform analysis to extract latent semantic features of images and combines this with an adaptive rate transmission mechanism, achieving a better balance

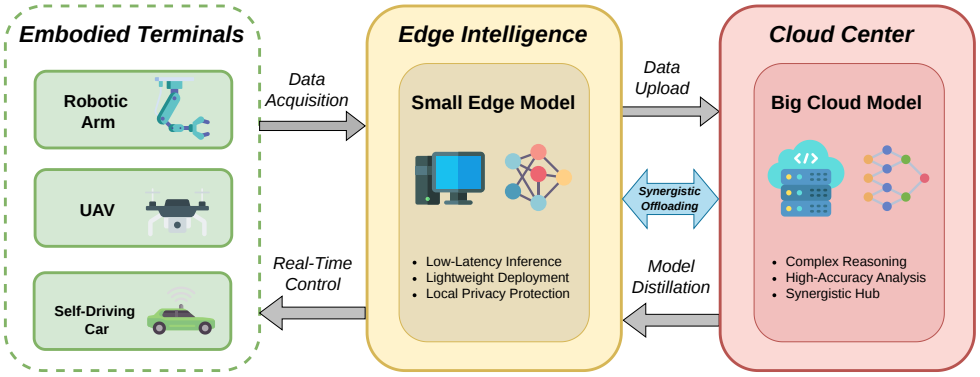

Fig. 6. Synergistic edge-cloud computing architecture for distributed perception and cognitive reasoning.

between end-to-end transmission distortion and perceptual quality. To further compress multi-scale semantic features, VQ-DeepSC [43] employs codebook quantization. For data-intensive tasks such as autonomous driving, researchers have proposed several improvements based on previous work. For instance, performing feature segmentation only on critical object images, such as vehicles, pedestrians, and obstacles, reduces transmission redundancy [122]. Chao et al. proposed a Goal-Oriented Communication (GO-COM) architecture and a Goal-Oriented Semantic Variational Autoencoder (GOS-VAE) [19]. In the context of common interaction scenarios in Embodied Multimedia, a multi-user semantic communication system (MU-DeepSC) [204] was introduced. The system supports the collaborative transmission of images and text, allowing the receiver to extract corresponding modal features from different transmitters based on task requirements, thereby addressing task-oriented communication challenges in multi-user scenarios.

5.2 Edge-Cloud Computing

Embodied Multimedia systems require agents to process massive multimodal sensory streams, such as high-definition video and haptic feedback, while simultaneously executing real-time semantic reasoning and motion control. Relying solely on centralized cloud computing often incurs prohibitive latency and bandwidth fluctuations, rendering it unsuitable for the strict real-time constraints of Embodied Intelligence. Conversely, edge devices face inherent limitations in power consumption and computational capacity, restricting their ability to host large-scale multimedia models. Consequently, Edge-Cloud Computing (ECC) has emerged as a pivotal architecture, as illustrated in Figure 6, utilizing hierarchical deployment and synergistic scheduling of computational resources to sustain the continuous evolution of Embodied Multimedia.

Distributed Collaborative Architecture: The paradigm of centralized cloud computing is evolving toward End-Edge-Cloud Computing (EECC) to accommodate the complex demands of distributed AI in embodied systems. Under the framework, heterogeneous computational capabilities are orchestrated as an organic whole, where terminal devices handle lightweight feature extraction and real-time control, edge nodes manage local data analysis, and the cloud undertakes global model training [36]. To unify real-time data streams with AI tasks, the Cloud-Fog Embodied AI framework has been proposed to provide automated services closer to data sources, effectively bridging the gap between local and cloud computing [59]. Furthermore, unified edge-cloud migration frameworks facilitate the offloading of AI inference to the edge while retaining robust cloud processing for non-real-time large-scale analysis, a capability essential for collaborative robotics and video analytics [179]. Recent surveys further highlight that ECC has become the

fundamental paradigm for modern intelligent applications, particularly deep learning and LLMs, enabling systems to optimize the trade-off between performance, energy efficiency, and latency through distributed resource integration [108].

Synergistic "Big Cloud - Small Edge" Models: With the proliferation of Generative AI (GenAI) and Large Vision Models (LVMs), deploying these resource-intensive models on constrained embodied devices presents a significant challenge. A prominent solution involves a synergistic "Big Cloud Model and Small Edge Models" architecture, in which the cloud hosts large-scale foundation models for complex reasoning, whereas the edge executes lightweight models optimized via distillation for immediate perception [173]. Building upon the concept, NetGPT orchestrates cloud and edge LLMs to deliver hyper-personalized generative services, leveraging edge-side location awareness to optimize prompt completion and response velocity [26]. In industrial IoT scenarios, the LAECIPS framework demonstrates how large vision models assist edge-side adaptive collaboration by routing complex defect identification tasks to the cloud while processing routine inspections locally, thereby balancing high accuracy with low latency [60]. Additionally, to support high-throughput embodied data exchange, Near-Field Communication (NFC) has been integrated into Embodied Edge Intelligence (EEI), utilizing massive antenna arrays to achieve high spectral efficiency and secure communication for edge-side model support [90].

Federated Foundation Models and Privacy Preservation: As embodied agents interact with environments, they generate massive volumes of multimodal data involving user privacy, necessitating mechanisms to evolve models securely. Multi-Modal Multi-Task Federated Foundation Models (M3T-FFMs) have emerged as a solution, combining the generalization capabilities of foundation models with the privacy-preserving nature of Federated Learning (FL) to enable collaborative training at the wireless edge despite data heterogeneity and bandwidth constraints [14]. For specific human-robot interaction tasks such as electromyography (EMG)-based gesture recognition, researchers have developed enhanced federated edge learning models that utilize algorithms like FedAvg for distributed training, significantly reducing communication overhead while maintaining high recognition precision through model quantization [225].

Scenario-Specific Resource Optimization: The application of ECC in Embodied Multimedia must be rigorously optimized against the physical constraints of specific scenarios. In the context of Industry 5.0, human-centric smart manufacturing requires Cyber-Physical-Social Systems (CPSS) to possess high collaborative capabilities, utilizing cloud-edge synergy to achieve semantic fusion of manufacturing resources and autonomous evolution [208]. For instance, the Circular Embodied Intelligence Manufacturing (CEIM) architecture integrates multi-source perception with feedback mechanisms, employing large models to enhance environmental interaction and resolve the traditional separation between perception and execution [173]. In the Low Altitude Economy, Unmanned Aerial Vehicles (UAVs) require the integration of Sensing, Communications, Computation, and Control (ISC3), where Edge AI frameworks optimize offloading and flight control to enhance logistics efficiency [215]. Furthermore, addressing the immense energy consumption of high-performance computing in autonomous driving, Approximate Edge AI techniques have been proposed to process massive sensor data streams, finding an optimal equilibrium between algorithmic precision and resource expenditure [75].

6 COGNITIVE LAYER

The Cognitive Layer transforms multimodal perceptual inputs into physical actions. It redefines traditional paradigms: static retrieval evolves into dynamic agent-based exploration, and generative technologies shift from human-centric content creation to generating environmental dynamics, task plans, and executable actions. This section details

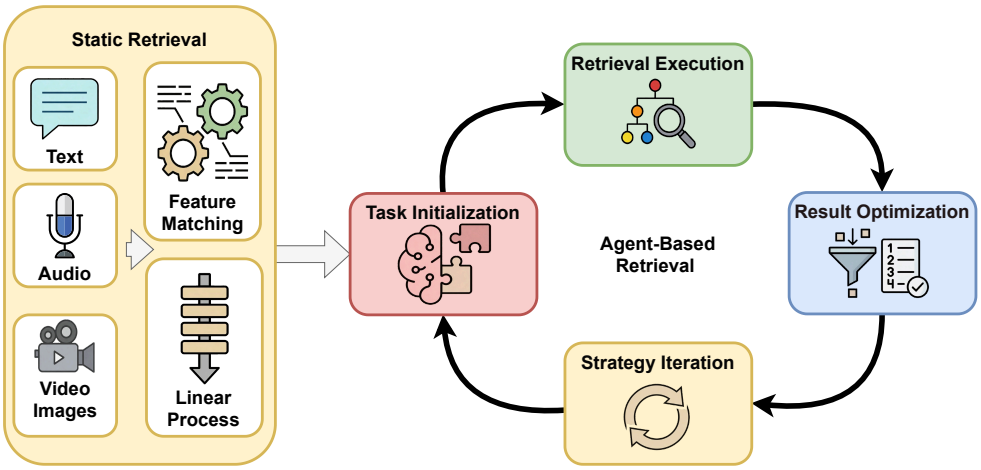

Fig. 7. The evolutionary transition from static matching to a dynamic, closed-loop agent-based retrieval paradigm, comprising four key optimization phases: task analysis, retrieval execution, result optimization, and strategy iteration.

how knowledge transfer, environmental prediction, and action decision-making address long-horizon planning in unstructured environments.

6.1 Agent-based Retrieval

Traditional cross-modal retrieval research primarily adheres to a static retrieval mechanism, constructing a unified feature representation space to achieve modal alignment by calculating similarities between heterogeneous data types, such as images and text. Classic works in the domain include statistical learning methods based on Canonical Correlation Analysis (CCA) [140], the Visual-Semantic Embedding (VSE) family of models [38, 42] and CLIP[137]. However, the static retrieval mechanisms lack the multi-step reasoning capabilities required for complex, knowledge-intensive embodied tasks. Consequently, researchers have shifted focus toward dynamic retrieval paradigms. This survey defines such frameworks—possessing capabilities for autonomous task analysis, retrieval strategy selection, and iterative optimization—as agent-based retrieval methods. The paradigm typically establishes a closed loop comprising task perception, strategy decision-making, and feedback-driven optimization, enabling the system to acquire and integrate external knowledge more efficiently in complex scenarios. The execution process is categorized into four key phases: task analysis, retrieval execution, result optimization, and strategy iteration, as illustrated in Figure 7.

Task Analysis Phase: During the task analysis phase, the agent first conducts semantic understanding and structural analysis of the retrieval requirements, determining subsequent strategies based on task complexity and modal characteristics. For ambiguous queries, decomposition into clear sub-queries is often adopted to mitigate semantic ambiguity. [76] explored a hierarchical processing flow of "query decomposition–sub-query retrieval–result aggregation" to improve semantic consistency, while [18] further optimized strategies regarding sub-query coverage and matching quality. For tasks involving multimodal information, such as text-to-image or text-to-video generation, the agent typically invokes retrieval models trained for multimodal semantic alignment to achieve representation mapping between modalities. For instance, CLIP aligns text and image embedding spaces through joint training [137], while CLAP extends the concept to text and audio [201]. In knowledge-intensive tasks like knowledge base question answering

Table 5. Comparison of generative and representation-prediction paradigms in world models.

Category	Year	Works	Core Contributions and Application Scenarios
Generative World Models	2023	VideoPoet[79]	**Contribution:** Proposed a Decoder-only Transformer architecture to handle multimodal inputs via generative objectives. **Application:** Zero-shot video generation and high-fidelity audio-video synchronization.
	2024	Genie[16]	**Contribution:** Created the first unsupervised generative interactive environment trained on unlabeled internet videos. **Application:** Generating controllable virtual worlds and training generalist agents.
	2024	WorldDreamer[185]	**Contribution:** Modeled world dynamics as an unsupervised visual sequence task via masked token prediction to internalize visual priors. **Application:** Text/Image-to-video generation, video editing, and physical simulation.
	2024	Drive-WM[186]	**Contribution:** Performed spatiotemporal modeling via view decomposition to generate high-fidelity multi-view driving videos. **Application:** Safety planning and optimal trajectory prediction via visual rewards.
	2025	DriveDreamer2[238]	**Contribution:** Integrated LLMs to convert user queries into trajectories and used a unified multi-view model for consistency. **Application:** Custom autonomous driving video generation and data augmentation for perception models.
	2025	Navigation World Models[11]	**Contribution:** Built a controllable video generation model using Conditional Diffusion Transformer (CDiT) to predict future observations. **Application:** Visual navigation planning and evaluating candidate trajectories in unfamiliar environments.
Representation-Predictive World Models	2025	V-JEPA 2[5]	**Contribution:** Extended JEPA to video by predicting masked spatiotemporal regions in feature space to learn semantics without pixel redundancy. **Application:** Video understanding and physical robot manipulation planning.
	2023	TD-MPC2[55]	**Contribution:** Improved model-based RL algorithm performing local trajectory optimization in latent space without pixel decoding. **Application:** Continuous control tasks across diverse domains and robust robot control.
	2025	DreamerV3[54]	**Contribution:** Developed a general world model algorithm using fixed hyperparameters to predict dynamics in latent space. **Application:** Mastering diverse tasks like Minecraft, Atari games, and robotic control.
	2023	DayDreamer[198]	**Contribution:** Applied the Dreamer architecture to physical robots enabling online learning of latent dynamics. **Application:** Quadruped walking, manipulator operation, and mobile robot navigation.
	2024	OccWorld[241]	**Contribution:** Built a world model in 3D Occupancy space using a GPT-like Transformer to predict scene evolution. **Application:** Scene evolution prediction and map-less motion planning for autonomous driving.
	2024	DriveWorld[127]	**Contribution:** Proposed a 4D scene latent dynamics pre-training framework combined with a dynamic memory bank. **Application:** Autonomous driving 3D detection, tracking, and motion prediction.

(KBQA), the agent can initiate structured knowledge retrieval processes. RNG-KBQA proposed a k-hop neighbor retrieval paradigm [220], and TIARA introduced a multi-grained knowledge extraction mechanism [157], providing explicit structured knowledge support for subsequent modules.

Retrieval Execution Phase: During the retrieval execution phase, agent-based retrieval methods primarily focus on combining retrieval paradigms and dynamically scheduling retrieval units. Regarding retriever selection, existing research widely adopts a collaborative mechanism between sparse and dense retrievers: sparse retrievers, represented by BM25, excel at term-level exact matching [142], while dense retrievers, represented by DPR, capture deeper semantic correlations [74]. The hybrid design has been validated in works such as RAP-Gen [182], BlendedRAG [147], and ReACC [118]. In terms of retrieval unit optimization, agents can dynamically adjust retrieval granularity by combining block-level optimization and hierarchical organization strategies. [107] proposed first locating key sentence blocks and then expanding the context window to balance retrieval precision with context completeness. [145] introduced a multi-level tree construction method, RAPTOR, which organizes text data into a hierarchical tree structure through recursive embedding, clustering, and summarization, enabling adaptive adjustment of retrieval granularity. For complex tasks requiring multi-step information exploration, retrieval actions are often triggered dynamically by the reasoning process. For instance, the ReACT framework determines when to invoke retrieval tools by alternating between reasoning

and acting [217]. The Chain-of-Thought method explicitly expands the intermediate decision-making process through step-by-step reasoning [193], while RATP models the selection of retrieval strategies as a tree search problem [135].

Result Optimization Phase: Research in the result optimization phase concentrates on enhancing the quality of retrieved content. Common techniques include result reranking, content transformation, and retriever adaptation. Regarding reranking, model-based relevance assessment methods are employed to re-evaluate the importance of candidate texts [41]. Additionally, some studies assess the influence of retrieved content based on generation bias and optimize ranking strategies accordingly [139]. Both approaches effectively reduce noise interference, adhering to the "retrieve-rerank-generate" pipeline validated by Re2G [46]. In terms of content transformation and retriever adaptation, existing research primarily improves the compatibility and computational efficiency of retrieved results for generative models through information filtering, structural reorganization, and feedback-driven parameter adjustments. Specifically, FILCO and FiD-Light optimize retrieval context via redundant content filtering and representation compression, respectively [58, 187]. RRR improves input structure through template-based integration [4], while REPLUG [153] and APICoder [226] leverage generation feedback or domain-specific data to continuously adapt the retriever, enhancing its efficacy within specific task semantic spaces.

Strategy Iteration Phase: When initial retrieval results fail to fully meet task requirements, the agent can initiate a re-retrieval process based on generative feedback, forming a dynamic optimization loop of "evaluation–adjustment–re-retrieval." CRAG proposed a mechanism for dynamic switching between different retrieval sources [211] to enhance system adaptability across various scenarios. At the parameter optimization level, the agent dynamically adjusts retriever hyperparameters based on quality metrics provided by the generation module—such as information relevance and factual accuracy—thereby continuously optimizing the alignment between retrieval results and task objectives.

Agent-based retrieval methods demonstrate significant advantages in dynamism, interpretability, and task adaptability. The approach adjusts retrieval behavior in real-time based on context, reducing interference from redundant information. Furthermore, it achieves traceability and transparency in the retrieval decision-making process through internal reasoning chains, marking a major advancement in Embodied Multimedia retrieval.

6.2 World Model

By learning the inherent laws governing how environmental states evolve over time, world models enable embodied agents to "imagine" the future and infer consequences, providing crucial support for decision-making and planning. Existing world models can be classified into two categories based on their prediction targets: generative world models and representation-prediction world models, as summarized in Table 5. Two representative world models are depicted in Figure 8.

Generative World Models: The category aims to model environmental dynamics by generating high-fidelity future images and video frames [79] or by creating interactive environments that adhere to physical laws [16]. For instance, WorldDreamer [185] demonstrates the potential for zero-shot video generation driven by large-scale data, suggesting that the model has internalized extensive visual prior knowledge. In autonomous driving scenarios, studies have extended generative world models to multi-view video generation [186, 238]. In navigation tasks, Navigation World Models [11] employ conditional diffusion Transformers to build controllable video-generative world models. The models simulate environmental dynamics and evaluate candidate trajectories, enabling effective planning in unfamiliar environments by leveraging visual priors.

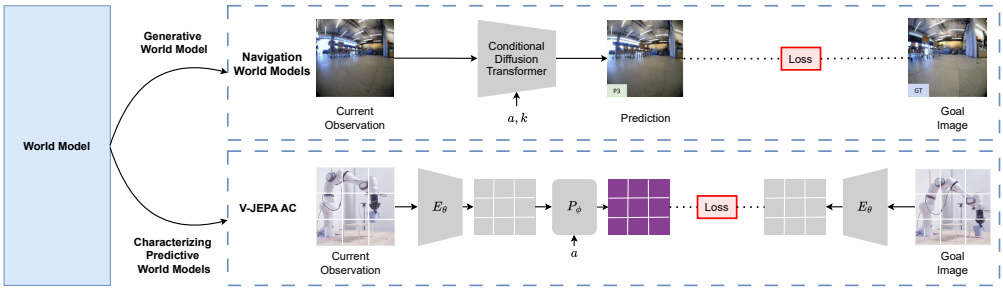

Fig. 8. Illustration of two representative world models: Navigation World Models[11] and V-JEPA AC[5].

Table 6. Comparison of environment model-based and reasoning-based vision-language navigation paradigms.

Category	Method	Year	Key Mechanism	Main Contribution
Environment Model-based	DUET[22]	2022	Topological Map	Captures global structure, enhances long-term memory
	MapGPT[21]	2024	Online Language Map	Builds explicit spatial representation, captures global structure
	VER[111]	2024	Voxel Reconstruction	Fine-grained physical reconstruction and representation
	Lookahead[191]	2024	Neural Radiance Field	Environment prediction and lookahead path tree search
	NavMorph[218]	2025	Self-evolving World Model	Simulates dynamic environments, enhances prediction
	WMNav[131]	2025	Curiosity Value Map	Introduces curiosity mechanism for active exploration
Reasoning-based	NavGPT[243]	2024	Text-based Reasoning	Decomposes task into logical deduction, reducing map dependency
	PixNav[17]	2024	Semantic Planning	Semantic planning guides low-level control
	NaVid[227]	2024	Video-based VLM	Simulates intuitive perception, map-free end-to-end decision
	Discuss Before Moving[115]	2024	Multi-expert Debate	Multi-view logical consensus, overcoming single-model limitations
	SmartWay[154]	2025	History-aware Backtracking	Long-term error correction, improving system robustness
	NaviLLM[240]	2024	Unified Paradigm	Unified paradigm, advancing VLN towards AGI

Representation-Prediction World Models: Unlike models that directly generate high-dimensional pixel data, representation-prediction models predict the future within an abstract latent or feature space. They focus on understanding essential environmental characteristics and planning rather than generating realistic future imagery. As the methods generally do not require decoding data into human-readable images, they offer significant advantages in computational efficiency and robustness against environmental noise. Building upon I-JEPA [6], V-JEPA 2 [5] learns semantically rich visual models by predicting masked feature representations, thereby avoiding the redundancy associated with pixel-level prediction. In motion planning and control, TD-MPC2 [55] and the Dreamer model series [52–54] (including DayDreamer [198] for physical robots) demonstrate how to perform efficient planning within latent dynamics models to master diverse control tasks. Furthermore, the approach has been extended to autonomous driving perception and planning; for instance, OccWorld [241] predicts 3D occupancy grids, while DriveWorld [127] focuses on understanding latent dynamics in 4D scenes. Collectively, the studies demonstrate that by predicting abstract future representations, models can focus more on task-relevant state changes, leading to more efficient reasoning and decision-making.

6.3 Vision-Language-Navigation

VLN [3] mandates that agents accurately translate natural language instructions into sequences of navigation actions grounded in visual signals within unstructured real-world or continuous environments. The VLN task has emerged as a core benchmark for evaluating the collaborative capabilities of perception and cognition in embodied agents. Current research in VLN is primarily categorized into two paradigms: environment model-based and reasoning-based approaches, as summarized in Table 6.

Environment Model-based: Methods based on environment models emphasize constructing explicit spatial representations to enhance the long-term memory and planning capabilities of agents. Among the methods, DUET

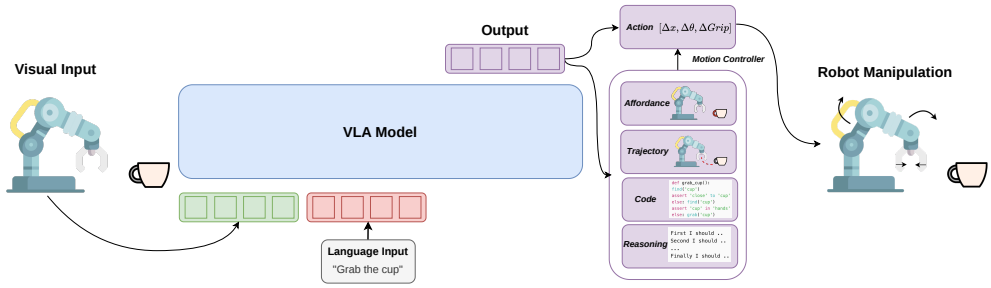

Fig. 9. Vision-language-action models mapping multimodal inputs to diverse physical execution representations.

[22] and MapGPT [21] capture the global structure of the environment by establishing topological maps or online language maps. Meanwhile, VER [111] and Lookahead Exploration [191] concentrate on detailed physical environment reconstruction and path tree searches using voxels or Neural Radiance Fields (NeRF). NavMorph [218] and WMNav [131] further introduce self-evolving world models or curiosity-driven value maps to simulate dynamic environments, significantly strengthening the predictive capabilities of agents.

Reasoning-based: In contrast, reasoning-based methods leverage the general cognitive capabilities of foundation models for inference and decision-making, thereby reducing reliance on complex geometric maps. For example, NavGPT [243] utilizes LLMs to decompose navigation tasks into logical deduction processes executed via textual reasoning. PixNav [17] employs the semantic planning of large models to guide low-level control, effectively bridging the gap between abstract language and concrete actions. Building upon the preliminary benchmarks, NaVid [227] further utilizes video-based Vision-Language Models (VLMs) to mimic human intuitive perception, achieving end-to-end decision-making without depending on external maps or odometry. To overcome the cognitive limitations of individual models, Discuss Before Moving [115] and SmartWay [154] introduce multi-expert collaborative debate and history-aware backtracking mechanisms, respectively. The approaches significantly improve system robustness through multi-perspective logical consensus and long-term error correction. Finally, NaviLLM [240] constructs a unified general navigation paradigm, marking a critical step for VLN toward artificial general intelligence.

6.4 Vision-Language-Action

Serving as the decision-making hub, VLA models map high-level semantic understanding to low-level robotic control. As shown in Figure 9, they directly output executable actions based on foundation models, or indirectly generate actions via affordances, trajectories, code, or reasoning. This end-to-end paradigm has established several distinct technical pathways across robotic manipulation and autonomous driving.

Action Token as Output: Mainstream VLA models adopt an "Action Tokenization" strategy, representing robotic actions as tokens within a unified structure alongside text and vision. The strategy allows Transformer-based architectures to achieve perception and control within a single sequence space. RT-1 [15] pioneered the validation of this paradigm on large-scale real-world robotic data, demonstrating the strong scalability of Transformers in multi-task robot learning. RT-2 [246] further integrated action tokens directly into the vocabulary of large VLMs and performed joint fine-tuning with internet-scale vision-language data, thereby enabling the transfer of semantic knowledge to robotic manipulation capabilities. The research suggests that high-level semantic structures inherent in large models can be harnessed for embodied tasks, providing new possibilities for complex instruction understanding and zero-shot generalization. To address the challenge of acquiring large-scale labeled data, LAPA [219] learns latent actions from

unlabeled internet videos, reducing dependence on robotic action labels and expanding the data sources for training VLA models. To mitigate model opacity and improve fine-tuning efficiency, OpenVLA [78] integrates general language models with vision encoders, proving the feasibility of efficient fine-tuning on consumer-grade hardware. At the architectural level, π_0 [13] introduces a Flow Matching mechanism to replace traditional discrete token prediction; combined with pre-trained VLMs, it achieves a cross-platform general robotic control policy. Addressing the challenge of high-frequency action representation, FAST [134] subsequently proposed a frequency-domain tokenization scheme based on the Discrete Cosine Transform (DCT), validating its significant advantages in training efficiency and dexterous manipulation when combined with the π_0 architecture. Finally, OpenVLA-OFT [77] proposed an optimized fine-tuning strategy involving parallel decoding for action representation, substantially increasing the inference throughput and task success rate of VLA models.

Affordance and Trajectory as Output: A subset of VLA models aligns the abstract semantics in language models with the physical environment by outputting affordances or trajectories to provide guidance for low-level controllers. SayCan [66] balances semantic plausibility with physical feasibility by combining high-level task suggestions generated by LLMs with value function evaluations. CLIPort [155] employs a two-stream architecture that decouples "What" to do from "Where" to do it, achieving high spatial precision in tabletop manipulation tasks. For more complex 3D scenes, VoxPoser [64] maps linguistic semantics into 3D value maps to provide spatial constraints for motion planning. Additionally, some methods introduce trajectories as intermediate representations. RT-Trajectory [49] supplements the motion information in language descriptions with trajectories, while ATM [194] predicts point-level trajectories at the video level. The aforementioned methods, to some extent, mitigate the inherent limitations of language in describing fine-grained movements. Similar concepts have been applied in autonomous driving; for example, DriveVLM [172] combines vision-language understanding with hierarchical planning to output real-time trajectory planning for handling complex traffic scenarios.

Code and Reasoning as Output: In addition to the aforementioned approaches, some VLA studies utilize code or reasoning as intermediate representations to bridge semantics and execution. Code as Policies [98] leverages LLMs to translate natural language instructions into executable programs, enabling agents to explicitly call functions and tools, which enhances policy compositionality and interpretability. ProgPrompt [158] further constrains the generation process through programmatic prompting, exhibiting superior stability in real-world robotic systems. To support long-horizon task execution, Inner Monologue [65] introduces a closed-loop mechanism based on environmental feedback. The feedback mechanism converts perceptual results into text and feeds them back into the model, allowing the policy to adjust dynamically based on execution outcomes. The methods emphasize the continuous interaction between perception, reasoning, and action, providing a higher level of flexibility for complex tasks.

7 EVALUATION LAYER

The Evaluation Layer establishes a comprehensive assessment framework for perception, cognition, and task execution. As depicted in Figure 10, this section delineates the multi-dimensional evaluation methodologies for Embodied Multimedia systems across three dimensions: perception evaluation, cognitive evaluation, and action evaluation.

7.1 Perception Evaluation:

Traditional image and video evaluation metrics are typically designed based on the human visual system. However, with machines emerging as the primary consumers of visual data, researchers have proposed new assessments for machine-perceived quality. Li et al. [88] identified a significant divergence between machine and human preferences,

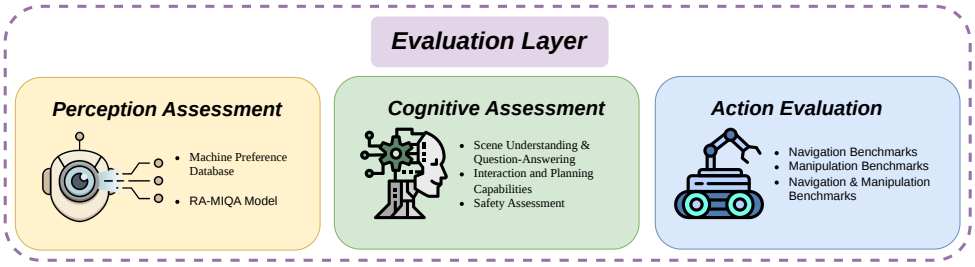

Fig. 10. Multi-dimensional evaluation framework encompassing perception, cognition, and action assessment.

introducing the core concept of "Machine Preference." The proposed concept defines image quality strictly by the performance consistency of downstream tasks, such as object detection and semantic segmentation. By constructing the Machine Preference Database (MPD), comprising 2.25 million finely annotated samples, the team demonstrated that traditional human-centric evaluation metrics cannot accurately characterize the perceptual requirements of machines. Consequently, they established a machine perception scoring standard based on task efficacy. Building on this, Wang et al. [190] further refined evaluation metrics by proposing the Region-Aware Machine Quality Assessment (RA-MIQA) model. The model evaluates the quality of machine vision systems through fine-grained spatial degradation analysis and reveals the sensitivity of machines to specific regions—such as those exhibiting background degradation—thereby providing more precise support for machine vision tasks.

7.2 Cognitive Evaluation:

Cognitive evaluation encompasses multiple dimensions, ranging from basic scene understanding and question answering to agent interaction, planning capabilities, and safety assessment, with the objective of comprehensively quantifying the cognitive maturity of embodied agents. In the realm of embodied question answering and spatial-physical understanding, Embodied Question Answering [32] pioneered the task paradigm integrating visual navigation with linguistic question answering, mandating that robots explore environments to answer questions. Subsequently, ScanQA [7] extended the paradigm to 3D point cloud scenes, focusing on the fine-grained association between language instructions and 3D geometric features as well as object localization capabilities. SQA3D [123] requires agents to comprehend the environment and perform reasoning based on their specific position and orientation within a 3D scene. More recently, PhysBench [29] incorporates physical laws, evaluating the capacity of vision-language models to discern object attributes, physical relationships, and dynamics through video-image-text data. Regarding the assessment of interaction and planning capabilities, SEAN 2.0 [176] introduces complex pedestrian dynamics generated by behavior graphs to specifically evaluate a robot's ability to navigate and interact while adhering to social norms in populated scenarios. EmbodiedEval [27] constructs a unified evaluation framework encompassing navigation, object manipulation, social interaction, and question answering. EMMOE [89] targets mobile manipulation in open environments, primarily testing the instruction-following and replanning capabilities of agents when executing long-sequence daily tasks. In the area of safety assessment, IS-Bench [119] employs interactive environments to evaluate the procedural safety of agents in sequentially executing mitigation measures upon perceiving risks. AGENTSAFE [103] utilizes adversarial scenarios to specifically test the defense capabilities and compliance of agents when confronted with dangerous or malicious instructions.

Table 7. Summary of major datasets and benchmarks for action evaluation.

Dataset/Benchmark	Year	Simulator	Continuous/Discrete	Purpose
R2R[3]	2018	Matterport3D	Discrete	Navigation
R4R[67]	2019	Matterport3D	Discrete	Navigation
RxR[80]	2020	Matterport3D	Discrete	Navigation
REVERIE[136]	2020	Matterport3D	Discrete	Navigation
ALFRED[156]	2020	AI2-THOR	Discrete	Navigation & Manipulation
BEHAVIOR[161]	2022	iGibson	Continuous	Navigation & Manipulation
OVMM[221]	2023	Habitat	Continuous	Navigation & Manipulation
RLBench[68]	2020	CoppeliaSim	Continuous	Manipulation
ManiSkill[128]	2021	SAPIEN	Continuous	Manipulation
CALVIN[125]	2022	PyBullet	Continuous	Manipulation
LIBERO[104]	2023	MuJoCo	Continuous	Manipulation
Open X-Embodiment[132]	2024	–	Continuous	Manipulation

7.3 Action Evaluation:

The domain of action evaluation encompasses several distinct benchmarks covering a wide spectrum of task scenarios, from navigation to complex manipulation, as summarized in Table 7, collectively driving the advancement of Embodied Intelligence. R2R [3] represented the inaugural VLN benchmark based on real-world environmental images. R4R [67] extended the trajectories, mandating that agents possess long-distance navigation capabilities. RxR [80] introduced multilingual support and dense spatio-temporal alignment annotations, emphasizing the fine-grained grounding of language and visual signals during navigation. Building on these, REVERIE [136] further increased the difficulty by requiring robots to navigate to unseen environments and identify specific objects based on natural language instructions, focusing on remote embodied visual referring capabilities. ALFRED [156] concentrates on action learning for daily household tasks, significantly bridging the gap between simulation benchmarks and real-world applications by introducing long-sequence, compositional, and irreversible tasks. BEHAVIOR [161] encompasses 100 complex daily household activities, establishing a holistic system from activity definition to evaluation. OVMM [221] further challenges open-vocabulary mobile manipulation, requiring robots to navigate unseen environments and perform pick-and-place tasks with novel objects. RLBench [68] provides 100 unique, scalable, hand-designed tasks equipped with infinitely generated demonstration data, making it particularly suitable for few-shot and multi-task learning research. ManiSkill [128] utilizes a full-physics simulator to focus on training general manipulation skills through large-scale, high-quality demonstration data, paying special attention to the impact of object topology and geometric diversity on generalization. CALVIN [125] is dedicated to policy learning under long-horizon language conditions, evaluating the zero-shot generalization of models when facing new instructions and environments. LIBERO [104] includes 130 sequential tabletop manipulation tasks, focusing on action execution in lifelong learning scenarios. Finally, Open X-Embodiment [132] underscores the immense potential for training general robotic policies and achieving experience transfer across different hardware platforms by integrating large-scale datasets from 21 institutions and 22 different robot types worldwide.

8 DISCUSSION

This survey has systematically reviewed the unified architecture of Embodied Multimedia systems, covering the full pipeline from multimodal data acquisition and communication to cognitive decision-making and evaluation feedback. While recent advances in semantic communication, world models, and VLA models have substantially expanded the capability boundaries of embodied systems, existing solutions remain insufficient to support truly general-purpose,

robust, and real-time Embodied Multimedia intelligence. In this section, we synthesize the key research challenges identified across the literature and discuss emerging research directions.

8.1 Research Challenges

- **Multimodal Spatiotemporal Alignment:** Embodied agents operate within data environments that are inherently more complex than those encountered in conventional multimedia systems. The pronounced heterogeneity among sensing modalities presents significant challenges for spatiotemporal alignment. In addition to RGB imagery, embodied systems rely heavily on tactile data and LiDAR point clouds, which differ substantially in terms of sampling rates, signal structures, and physical characteristics. Consequently, achieving reliable alignment across these diverse modalities at a millisecond-level temporal resolution remains an open problem[40].
- **Physical Semantic Gaps in Data Representations:** Beyond modal alignment, profound semantic gaps persist at the data level. Most existing embodied datasets lack task-relevant physical attributes, such as friction coefficients, mass distribution, and material compliance[48]. As a result, models trained on purely perceptual data often lack access to the physical semantics required for robust manipulation and interaction.
- **Scarcity of High-Quality Interactive Data:** The scarcity of active and interaction-rich embodied data further limits the generalization capability of models. Unlike large-scale passive video corpora, datasets capturing causal interactions, failure cases, and long-tail scenarios are costly to acquire and remain underrepresented, thereby constraining the acquisition of transferable physical knowledge[132].
- **Hardware Constraints for On-Device Inference:** Embodied multimedia systems impose stringent real-time requirements that fundamentally conflict with the computational demands of contemporary foundation models. While cognitive layers increasingly rely on large-scale VLA models [77, 134] and world models [11] to achieve open-world understanding, embodied platforms—such as mobile robots and aerial vehicles—operate under strict limitations regarding power, latency, and thermal management. The physical constraints often preclude sustained on-device inference.
- **Network Unreliability in Edge-Cloud Synergy:** To address on-device computational bottlenecks, edge-cloud offloading offers a partial solution [36, 173]. However, reliance on continuous offloading introduces vulnerabilities to network conditions. Communication instability, which is characterized by bandwidth fluctuations and packet loss, severely compromises control continuity and operational safety in dynamic environments.
- **Sim-to-Real Transfer and Physical Generalization:** The sim-to-real domain gap remains a primary deployment barrier. While simulation training is prevalent [124, 146], physics engines' limited fidelity severely hinders physical generalization. Furthermore, models trained in closed [15, 155] or semi-structured [172] environments lack robustness against open-world distribution shifts, making the encoding of transferable physical priors a critical unresolved challenge [104].

8.2 Future Research Directions

- **Multimodal Active Sensing and Generation:** To address data scarcity and physical semantic gaps, future research must prioritize active multimodal sensing and physics-informed data generation. Synthesizing high-fidelity interactive data, particularly encompassing tactile feedback and complex dynamic properties, is crucial for training robust embodied agents [132].
- **Lightweight and Efficient Embodied Models:** To enable sustained on-device inference under strict hardware constraints, developing lightweight embodied foundation models is imperative. Future research should focus

on efficient network architectures, model compression, and parameter-efficient fine-tuning [101, 109, 197]. The optimization allows resource-constrained edge devices to independently execute complex multimodal perception and real-time control, fundamentally mitigating the latency inherent in cloud reliance.

- **Adaptive Edge-Cloud Collaboration:** Edge-cloud synergy is evolving toward adaptive hierarchical architectures [173]. In the paradigm, lightweight models handle time-critical, high-frequency physical responses on-device, while larger foundation models support remote long-horizon reasoning. Developing seamless, context-aware routing mechanisms across the computational layers remains a vital open problem.
- **Physics-Aware World Models:** A prevailing trend signals a shift toward physics-aware world models that predict environmental dynamics. Future architectures should prioritize latent representations capturing essential object properties, spatial structures, and temporal evolution over photorealistic generation, enabling agents to foresee physical consequences efficiently [5, 54].
- **Causal Reasoning and Policy Generalization:** Beyond predictive modeling, integrating causal reasoning into VLA architectures is essential. Enabling agents to distinguish correlation from true physical causation will significantly improve sample efficiency and policy robustness, thereby bridging the Sim-to-Real gap in unstructured, open-world scenarios [104].
- **Task- and Safety-Centric Evaluation:** There is an urgent necessity for standardized, multi-dimensional evaluation metrics. Future benchmarks must explicitly quantify not only task success rates but also continuous decision consistency, human-robot interaction safety, and hardware transferability across diverse embodied scenarios to systematically assess algorithmic progress [27, 89, 103, 119].

9 CONCLUSION

Motivated by the expanding reach of AI from digital environments into the physical world, this survey explicitly defines Embodied Multimedia as a cross-disciplinary research concept. By tracing the evolutionary trajectories of multimedia computing and Embodied Intelligence, we analyze the structural mismatch between traditional human-centric multimedia paradigms and the "perception-decision-action" closed loop required by embodied agents, particularly regarding data semantic modeling, system architecture design, and evaluation mechanisms. To achieve the objective, this survey constructs a unified Embodied Multimedia architecture, comprising the Data, Communication, Cognitive, and Evaluation Layer, which delineates the key processes and functional divisions involved in mapping multimodal perceptual information to physical actions. Building upon the proposed framework, this survey synthesizes the core technical directions supporting the architecture and systematically discusses the critical challenges currently facing Embodied Multimedia research, including data heterogeneity, model generalization, and real-time system performance. Looking ahead, Embodied Multimedia is poised to play a foundational role in bridging the Sim-to-Real gap, propelling the continued evolution of agents toward robust multimodal perception and generalized physical interaction capabilities.

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